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SUPPLEMENTARY MATERIAL

The SST Covariant Completion: y From the Acceleration of the Time Leg

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Abstract

The quasi-static proof of the SST [1] left three points open: the covariant definition of y, which the archive writes as $|\nabla\Phi|^2/a_0^2$; F for timelike gradients, $y<0$; and the metric field equations, with the status of Φ's stress–energy. We settle all three with one definition. Let $u=e_0$ be the time leg of the tetrad and $a^\mu=u^\nu\nabla_\nu u^\mu$ its acceleration, and set $y=c^4a_\mu a^\mu/a_0^2$. This y is a spacetime scalar. Since $a\cdot u=0$, the acceleration is spacelike or zero, so $y\geq 0$ for every configuration and F needs no extension. In a static metric $-N^2c^2dt^2+h_{ij}dx^idx^j$ with $N=e^{\Phi/c^2}$, $c^2a_i=\partial_i\Phi$ exactly, so y reduces to the archive's $|\nabla\Phi|^2/a_0^2$, and Φ is the lapse potential of the metric, not a separate field. The SST, with the reading of Lemma 1 of [1], is then the action $(1/16\pi G)\sqrt{-g}\,[R-2f(a)]$ with $f=a_0^2[F(y)-y]$. This is the preferred-foliation form of Blanchet and Marsat [8, 9], and it gives $\mu_{\rm total}=1+f'(a)/2a=x/(x+1)$ exactly. Its static weak field gives $\Psi=\Phi$ and $\nabla\cdot[\mu\nabla\Phi]=4\pi G\rho$; the metric equations carry the TetB share as phantom density; the time-leg stress–energy is post-Newtonian; and the homogeneous background is exactly general relativity. We also state a limit the covariant form makes precise: f does not tend to a constant at large a, so on each mass's own time leg the action leaves a TetB tail $\to a_0$. The archive excludes that tail in the solar system and applies μ to the galaxy's net field. The covariant statement of that rule is that a is the acceleration of the coarse-grained time leg; deriving that from an action is open.

Keywords: scalar-tensor transition, covariant action, tetrad, time leg, acceleration, khronometric, preferred foliation, MOND, proof

Setting

The SST is the action [2, 3, 4]

$$S=\int d^4x\sqrt{-g}\Big[\frac{R}{16\pi G}-\frac{a_0^2}{8\pi G}F(y) - \tfrac{1}{4}e^{-2\Phi/c^2}F_{\mu\nu}F^{\mu\nu}+\mathcal{L}_{\rm m}\Big], \tag{1}$$

with $F(y)=y-2\sqrt{y}+2\ln(1+\sqrt{y})$ and $\mu(x)=x/(x+1)$, $x=\sqrt{y}$. The quasi-static proof [1] reads the action as the framework does (Lemma 1 there): in the static weak field the curvature term supplies the Newtonian part y of F, and the scalar term supplies $F(y)-y$. It derives the field equation, $\gamma=1$ and the galactic results from that reading. Its Scope listed three open points, and this note answers them.

The time leg. The tetrad's leg e_0 is unit timelike, $g(e_0,e_0)=-1$. We take it normal to GFD's time foliation (in the static case, the slices of constant Killing time), so $u=e_0$ is hypersurface-orthogonal. Its acceleration is $a^\mu=u^\nu\nabla_\nu u^\mu$.

Definition. $y\equiv c^4\,a_\mu a^\mu/a_0^2$.

Each identity below is checked exactly by the accompanying script, 16 checks in all, using sympy for every algebraic step [7].

Contributions and prior work. An action of the form $(1/16\pi G)\sqrt{-g}[R-2f(a)]$, built from the acceleration of a unit timelike field normal to a preferred foliation, is Blanchet and Marsat's modified gravity with a preferred time foliation [8, 9]. Its MOND phenomenology, and the conditions on f, are theirs. Related constructions are Sanders' Hořava-type MOND [10], generalized Einstein-aether MOND [11] and the khronometric theory [12]. What this note adds:

- the identification of the SST's y with the acceleration of the tetrad's time leg;
- the SST's own f, $f=a_0^2[F(y)-y]$, which gives the archive's $\mu=x/(x+1)$ exactly;
- the identification of Φ with the lapse;
- the strong-field limit of this f, and what it means for the archive's net-field rule;
- machine checks of every step.

Theorem 1 (covariance and positivity) *y is a spacetime scalar, and $y\geq 0$ for every configuration. So $F(y)$ is defined everywhere, and no branch for timelike gradients is needed.*

Proof. y is the contraction of the vector a^μ with itself, so it is a scalar. From $u\cdot u=-1$, $u_\mu\nabla_\nu u^\mu=0$, so $a\cdot u=0$: a is orthogonal to a timelike vector and is therefore spacelike or zero. Hence $a_\mu a^\mu\geq 0$. The script checks $g(u,u)=-1$ and $a\cdot u=0$ identically for a general

time-dependent lapse and conformal factor, and $a \cdot a \geq 0$ at 20 random points of a time-dependent metric. □

So open point 2 does not arise. An argument $\nabla_\mu \Phi \nabla^\mu \Phi$ of a scalar Φ can be negative; the time leg's acceleration cannot.

Theorem 2 (the static reduction)
For a static metric $-N^2c^2dt^2 + h_{ij}dx^i dx^j$, the time leg's acceleration is $a_0 = 0$, $a_i = \partial_i \ln N$ exactly. With $N = e^{\Phi/c^2}$,

$$c^2 a_i = \partial_i \Phi, \qquad y = \frac{h^{ij}\partial_i\Phi\,\partial_j\Phi}{a_0^2}, \tag{2}$$

and in the weak field, $h_{ij} = (1 - 2\Psi/c^2)\delta_{ij}$, $y = |\nabla\Phi|^2/a_0^2 + O(c^{-2})$.

Proof. Direct computation of $u^\nu \nabla_\nu u^\mu$ with $u = (Nc)^{-1}\partial_t$ (checked for general N and a conformal h). Substituting $N = e^{\Phi/c^2}$ gives (2), and the limit $c \to \infty$ gives the weak-field form (checked). □

So the archive's y is the static form of a covariant scalar. The field strength is the acceleration of the frame's time leg: what an observer at rest in the foliation measures on an accelerometer. In the static case $\Phi \equiv c^2 \ln N$ is the lapse potential of the metric. It is not a field added to the metric, which is the framework's reading (i) in [1], now stated exactly.

Theorem 3 (the covariant SST)
With y as defined, the SST read as in Lemma 1 of [1] is the action

$$S = \int d^4x\, \frac{\sqrt{-g}}{16\pi G}\big[R - 2f(a)\big] + S_{\rm EM} + S_{\rm m}, \qquad f = a_0^2\big[F(y) - y\big], \tag{3}$$

in units $c = 1$, $y = a^2/a_0^2$. Then:

1. *Blanchet and Marsat's $\chi = f'(a)/2a$ is $\chi = \mu - 1$, so $\mu_{\rm total} = 1 + \chi = \mu(a/a_0) = x/(x+1)$ exactly;*
2. *the curvature term supplies the y of F, and the time-leg term supplies $F - y$: $-2f/16\pi G = -(a_0^2/8\pi G)[F(y) - y]$;*
3. *in the static weak field, with minimal coupling, $\nabla^2(\Phi - \Psi) = 0$, so $\Psi = \Phi$ ($\gamma = 1$), and*

$$\nabla \cdot \big[\mu(|\nabla\Phi|/a_0)\nabla\Phi\big] = 4\pi G\rho. \tag{4}$$

Proof. (1) Differentiate $f(a) = a_0^2[F(a^2/a_0^2) - a^2/a_0^2]$; using $F'(y) = \mu(\sqrt{y})$ gives $f'/2a = \mu - 1$ (checked exactly). (2) is the definition. (3) By Theorem 2 the static weak-field Lagrangian of (3) is Lemma 1 of [1],

$$\frac{(\nabla\Psi)^2 - 2\nabla\Phi\cdot\nabla\Psi}{8\pi G} - \frac{a_0^2}{8\pi G}\big[F(y) - y\big] - \rho\Phi.$$

Its Euler–Lagrange equations are checked: the Ψ-equation is $\nabla^2(\Phi - \Psi) = 0$, and the Φ-equation on $\Psi = \Phi$ is (4). □

Equation (3) is the reading of Lemma 1 written as a covariant action. Read with F in place of $F - y$, the y would be counted twice, and the field equation would be $\nabla \cdot [(1+\mu)\nabla\Phi] = 4\pi G\rho$ [1].

Theorem 4 (the metric equations and Φ's stress–energy)
Varying (3) gives Einstein's equation with the time-leg stress tensor as an extra source, plus the foliation's own equation [8]. There is no separate scalar stress tensor, because Φ is the lapse. In the static weak field:

1. *the 00-equation is $\nabla^2\Phi = 4\pi G(\rho + \rho_{\rm ph})$, with $\rho_{\rm ph} = \nabla\cdot[(1-\mu)\nabla\Phi]/4\pi G$, the TetB share as phantom density (for a deep point mass, $\rho_{\rm ph} = \sqrt{GMa_0}/4\pi G r^2$);*
2. *the time-leg energy density is the gravitational field energy, $\sim g^2/8\pi G$, since $F - y = -y + \frac{2}{3}y^{3/2} + O(y^2)$ in the deep field. As a mass density it is $v^2/2c^2$ times $\rho_{\rm ph}$, where $v^4 = GMa_0$: about 2×10^{-7} at 200 km/s.*

Proof. (1) is (4) rearranged, and the deep point-mass value is checked for $g = \sqrt{GMa_0}/r$. (2) The series of $F - y$ and the ratio are checked exactly. □

This answers open point 3. "Φ is the metric's Newtonian potential" is Theorem 2 ($\Phi = c^2 \ln N$) together with Theorem 3(3) ($\Psi = \Phi$). Its energy enters the metric at post-Newtonian order, as the energy of any gravitational field does.

Theorem 5 (the cosmological background)
In a homogeneous, isotropic background the comoving time leg is geodesic, $a = 0$, so $y = 0$ and $F(0) = 0$. The background is exactly general relativity, and the SST acts on perturbations.

Proof. For $-dt^2 + a(t)^2 d\mathbf{x}^2$ with $u = \partial_t$, every component of $u^\nu\nabla_\nu u^\mu$ vanishes (checked), and $F(0) = 0$. □

Theorem 6 (the two limits of f)
In the deep field, $f = -a^2 + 2a^3/3a_0 + O(a^4)$: Blanchet and Marsat's MOND condition with $\Lambda_0 = 0$. In the strong field, $f = -2a_0 a + 2a_0^2 \ln(a/a_0) + O(1)$, which does not tend to a constant, their condition for recovering general relativity. So the TetB share $(1-\mu)g$ tends to a_0 as $x \to \infty$.

Proof. Both series are checked exactly, as is $\lim_{x\to\infty} a_0\mu(x) = a_0$. □

What this means for the solar system. Evaluated on each mass's own time leg, the action leaves a TetB tail $(1-\mu)g \to a_0$ around every star. At Saturn, $x \approx 5 \times 10^5$ and the tail is a_0 to two parts in 10^6 (checked). This is the per-mass tail that the archive records as excluded: a constant pull $\approx a_0$ at every planet, ruled out by perihelia, and a Sun-centred quadrupole, ruled out by Cassini ([5], §5.4; [6], Part 3). The archive therefore applies μ to the galaxy's net field, never to a star's own field. The covariant form makes that rule precise: a is the acceleration of the coarse-grained time leg, the frame of the galaxy's net field, and bound subsystems gravitate internally through R alone. In a solar system that net field is uniform to $\sim 10^{-12}$ and is removed by free fall [5]. Equation (3) with a taken from each mass's own time leg does not implement this. An action that does, for example by defining the coarse-graining covariantly, is the open step.

Scope
Proved here:

- the covariant y and its positivity;
- its exact static reduction, and Φ as the lapse;
- the covariant form (3) of the SST under Lemma 1, with $\mu = x/(x+1)$ exactly;
- $\gamma = 1$ and the field equation from it;
- the structure and size of the metric equations' extra source;
- the general-relativistic background;
- the two limits of f.

Not proved here:

1. *The net-field mechanism* (Theorem 6).
2. *Preferred-frame effects.* A preferred foliation breaks local Lorentz invariance. The PPN parameters α_1, α_2 and the khronon's dynamics and stability [12] are not computed for this f. In Blanchet and Marsat's model, general relativity is recovered well above a_0 because f tends to a constant there (the cosmological constant); this f does not, so these must be computed with the net-field rule in place.
3. *Signal speeds.* The action has no $(\nabla u)^2$ shear or expansion terms, so the curvature term carries the tensor modes. The scalar-sector propagation speed found in the archive assumed Φ as a covariant scalar and needs redoing with Φ as the lapse.
4. *The fourth term.* With $\Phi = c^2 \ln N$, the EM coupling $e^{-2\Phi/c^2}$ is defined relative to the foliation's normalized time; its coefficient α remains open, as in [1].
5. *A general time leg.* If e_0 is not hypersurface-orthogonal, the theory is of generalized Einstein-aether type [11], with a vorticity sector not treated here.

References

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